GRADUATION PROJECT

MODEL2PRODUCT AI

Project Planning & Management

AI Tech Lead Mentor & Project Delivery Simulator

8	4	MVP
WORKSTREAMS	TEAM MEMBERS	DELIVERY FIRST

A controlled, stage-gated delivery plan that prioritizes verifiable learning, safe implementation and a deployable end-to-end demo.

INITIATIVE

Digital Egypt Pioneers Initiative (DEPI)

DOCUMENT STATUS

Planning draft | v1.1

Prepared 16 October 2026

Owner / approval / start date: to be confirmed

02 / PROJECT CHARTER

Management brief

Flexible	4	5-6	3+
Timeline	Team members	Target core agents	Minimum demo agents

PROJECT PURPOSE

Close the gap between training an AI model and shipping a complete software product. The platform behaves as an AI Tech Lead: it directs the next stage, explains realistic trade-offs, checks learner understanding and only then authorizes implementation assistance.

BUSINESS AND EDUCATIONAL OUTCOMES

Learner impact

Improve engineering decision-making, not merely code output.

Instructor visibility

Expose decisions, learner justification, AI-generated artifacts and skill gaps.

Delivery impact

Produce usable requirements, architecture, schemas, APIs, tests and deployment evidence.

Product credibility

Demonstrate a stateful, safe, testable and deployed full-stack experience.

CHARTER CONTROL FIELDS

Sponsor / instructor	To be assigned
Project / technical lead	To be assigned
Calendar start date	To be confirmed
Submission deadline	To be confirmed with supervisor
Primary beneficiaries	AI students, junior AI engineers
Buyers / stakeholders	Universities, academies, bootcamps

PLANNING RULE

Items not explicitly fixed by the source proposal are proposed defaults and require team approval.

03 / BASELINE & SCOPE

Build the right MVP

MVP NORTH STAR

A learner completes a guided engineering journey, makes defensible decisions and receives verified implementation artifacts.

INCLUDED IN THE GRADUATION MVP

01 Guided stages

Project creation, discovery, stage locks, required decisions and explicit completion conditions.

02 Multi-agent mentoring

Tech Lead orchestrator and at least three demonstrated specialist roles sharing project state.

03 Grounded engineering advice

Curated knowledge base, source-aware RAG and structured option / trade-off cards.

04 Approved engineering outputs

Generate at least two artifacts (e.g., architecture diagram and ERD/API spec).

05 Safe implementation help

Create or modify a small scaffold inside an isolated workspace and run a check/test.

06 Measurable progress

Persist decisions and artifacts; show project progress and competency indicators.

DEFERRED / NOT REQUIRED FOR ACCEPTANCE

- Training a foundation LLM from scratch or shipping fine-tuned custom model weights.
- Autonomous full-product coding, unrestricted IDE or cloud deployment actions.
- Large enterprise analytics, organization-specific standards and full team mode.
- Support for every software domain and all 8-10 envisioned specialist agents.

SCOPE INTERPRETATION

Proposal vision: 5-6 core agents in the target MVP. Hard demo acceptance: at least 3 specialists, 4 consecutive guided stages, 2 decision comparisons and 2 generated artifacts. Do not confuse the broader product vision with the minimum passing slice.

04 / EXECUTION DESIGN

Work breakdown structure

Eight accountable workstreams organize delivery. Each begins when prerequisite work is approved; no fixed project duration is assumed.

WBS	WORKSTREAM	START AFTER	EVIDENCE / OUTPUT
1.0	Governance & discovery	Kickoff	Charter, backlog, stakeholder map, functional / non-functional requirements.
2.0	UX & architecture	Scope sign-off	User journey, approved architecture comparison, component diagram, data model.
3.0	Core platform & data	Design sign-off	Auth, project APIs, persistent stages, decision log, artifact repository.
4.0	Knowledge & evaluation	KB scope	Curated RAG corpus, retrieval pipeline, scenario set and evaluation rubric.
5.0	Agents & state gates	Core state/API	Tech Lead router, specialist agents, state rules, justification checkpoint.
6.0	Frontend & progress	UX sketch	Project workspace, stage map, comparison UI, progress / competency view.
7.0	Safe artifact tooling	Agent flow	Diagrams / API schemas, isolated code scaffold, automated tests, security checks.
8.0	Integration & release	Core integration	E2E tests, deployable system, monitoring, final docs and evaluated live demo.

BUILD VERTICAL SLICES, NOT ISOLATED AGENTS

First demonstrate one stage end-to-end: UI -> API -> state -> agent -> decision -> approved artifact -> saved evidence.

05 / FLEXIBLE ROADMAP

Delivery phases - no fixed weeks

This roadmap is dependency-based, not a calendar promise. Phases may overlap, and target dates will be agreed with the supervisor once the final deadline is known.

ACTIVE WORKSTREAM

REVIEW GATE



REVIEW CHECKPOINTS

Critical path: approve scope -> architecture -> shared state/RAG -> gated agents -> safe artifacts -> integration tests -> deployed demo. Gate dates are TBD.

06 / DELIVERY CONTROLS

Milestones & exit gates

A milestone is complete only with a reviewable artifact, a testable outcome and a recorded approver. Dates remain unset until the team agrees the deadline.

REVIEW GATE	EVIDENCE REQUIRED	ACCOUNTABLE TEAM
G0 | Charter Project charter	Approved problem statement, success criteria, baseline scope and open decisions.	M1 - Lead
G1 | Scope Requirements freeze	User stories, prioritized backlog, acceptance criteria and demonstrable 4-stage path.	M1 - Lead
G2 | Design Design sign-off	Architecture comparison, ADR, data schema/ERD, API boundary and UI flow.	M2 + M4
G3 | Platform Platform foundation	Working API, auth, persisted project state, knowledge ingestion and basic retrieval.	M4 + M3
G4 | Mentor flow Mentor vertical slice	At least 3 specialist agents; stage lock, 2 structured decisions and user justification.	M2 - Agentic
G5 | Tooling Artifact & tool proof	Two approved generated artifacts; isolated scaffold action plus passing automated check.	M4 + M1
G6 | QA Quality gate	End-to-end scenarios, security checks, regression results and readiness issues closed.	M4 - Backend
G7 | Deployment Deployed acceptance	Accessible deployed web app with persistent user flow, dashboard and runbook.	M4 + M1
G8 | Final Final submission	GitHub source, README, technical report, presentation and repeatable demo.	All 4 members

Gate authority: Tech Lead documents evidence; instructor / supervisor provides formal approval where required.

07 / PEOPLE & OWNERSHIP

Four-member responsibility plan

Exactly four members share the product, AI, backend and release responsibilities. Member labels are placeholders until names are confirmed.

M1 — **Product Lead + Frontend/UX**
Scope, architecture, UI, dashboards, handover

M2 — **Agentic AI + Orchestration**
Tech Lead router, agents, stage gates, memory

M3 — **RAG + Knowledge + Evaluation**
Retrieval, rubrics, benchmark cases, AI metrics

M4 — **Backend + DevOps + QA**
FastAPI, PostgreSQL, security, CI/CD, deployment

RACI | A = ACCOUNTABLE, R = RESPONSIBLE, C = CONSULTED

WORK PACKAGE	M1	M2	M3	M4
Scope & acceptance	A/R	C	C	C
Architecture / ADR	A/R	R	C	R
RAG quality & metrics	C	C	A/R	C
Orchestrator / stage gates	C	A/R	R	C
Platform & APIs	C	C	C	A/R
Frontend / progress	A/R	C	C	C
Testing / release / security	R	R	R	A/R
Final demo / handover	A/R	R	R	R

CROSS-REVIEW RULE

Because no fifth QA member exists, every high-risk change needs review by another member. M4 owns the release checklist; the whole team runs end-to-end acceptance.

08 / RESOURCES & ARCHITECTURE

Technology and operating plan

The stack below reflects the proposal recommendations; choices with alternatives should be formally selected by the team at design sign-off.

SYSTEM LAYER	PROPOSED CHOICE	DELIVERY / CONTROL NOTE
Client	React / Next.js + Tailwind	Stage map, mentor chat, dashboards
API services	Python + FastAPI	Typed contracts, authorization, validation
Workflow engine	LangGraph OR explicit state machine	Checkpointed transitions and agent control
Models & knowledge	LLM API or open model; embeddings + RAG	Source versioning, latency and cost tracking
Persistent data	PostgreSQL; pgvector or Qdrant	Projects, decisions, rubrics, retrieval
Files & tooling	Object storage; Mermaid; Docker sandbox	Versioned artifacts and isolated execution
Ops & delivery	GitHub, CI/CD, Docker / managed cloud	Audit trail, tests, deployment observability

BUDGET PLANNING: COST CATEGORIES (AMOUNTS PENDING)

LLM usage
Token budget, caching, fallback and hard spend caps

Cloud hosting
VM/container, database, object storage, egress

Data and testing
Synthetic evaluation cases, monitoring/traces

Team tooling
Repository, design, issue tracking and CI minutes

09 / PROACTIVE MANAGEMENT

Risk register & response

These risks come from the proposal. Risk ratings, owners and trigger examples are planning suggestions to validate at kick-off.

ID	RISK / WARNING SIGNAL	LEVEL	RESPONSE	OWNER
R01	Scope creep: added stages or integrations mid-sprint	HIGH	Freeze MVP at G1; move extras to stretch backlog.	M1
R02	Conflicting specialist advice / inconsistent state	HIGH	Structured outputs; centralized decisions; regression suite.	M2
R03	Stale or weak technical grounding	MED	Versioned RAG corpus, citations and reviewer sampling.	M3
R04	Latency / token spend beyond acceptable demo use	MED	Usage budgets, caching, smaller-model routes, async jobs.	M2 + M3
R05	Learners bypass checkpoints to obtain code	HIGH	Enforce state gates server-side; audit approvals.	M2
R06	Unsafe generated-code execution / secret exposure	HIGH	Docker isolation, permission limits, secrets policy.	M4 + M1
R07	Poor evidence that the mentoring approach works	HIGH	Pre/post evaluation and mentor-reviewed benchmarks.	M3
R08	End-to-end integration breaks near final demo	HIGH	Build an early vertical slice; weekly E2E regression.	M1 + M4
R09	Project files or state leak between user workspaces	HIGH	Authorization and data isolation tests before release.	M4 + M1

Risk rule: record risk owner, next action, target checkpoint and residual exposure at each status review.

10 / ASSURANCE PLAN

Testing, evaluation & KPIs

SIX COMPLEMENTARY VALIDATION LAYERS

01 Unit tests
Stage transitions, schemas, parsers, rubric logic

02 Integration
Agent -> state -> DB -> RAG -> tooling

03 LLM / RAG evaluation
Grounding, trade-offs, consistency, hallucinations

04 Security
Permissions, sandbox, injection, secret handling

05 User acceptance
Guided project path, explanation quality, progress

06 Performance / regression
Latency, workflow success and repeatability

MEASURED DASHBOARD (BASELINE METRICS FROM SOURCE)

MEASURE	EVIDENCE / INTERPRETATION
Stage completion / compliance	Completed eligible stages; no unlawful transition or missing approval.
Decision quality / learning uplift	Rubric-backed justifications plus pre/post competency checks.
Artifact quality / project readiness	QA-approved artifacts and checklist coverage across key layers.
Retrieval grounding / model reliability	Evidence relevance, correct sources and repeated-scenario consistency.
Latency / cost / instructor intervention	Observed response time, token use, and mentor rescue frequency.

RECOMMENDED QUALITY GATE

No unresolved critical security issue; all mandatory acceptance tests pass; reported metrics include method, test set and evidence. Numeric KPI thresholds must be approved by the team.

11 / PROJECT OPERATIONS

Cadence & change control

A lightweight two-week sprint rhythm is proposed for control, not prescribed by the source. Maintain a single backlog and an immutable decision trail.

OPERATING CADENCE

Every week	**Execution check**	Progress, blockers, risks, cost and dependency check (20-30 min).
Every 2 weeks	**Sprint planning / review**	Demo working slice, re-order backlog and capture instructor feedback.
At major gates	**Artifact approval**	Sign-off on requirements, architecture, mentor flow, QA and release.
Before final demo	**Readiness rehearsal**	Full acceptance scenario, fallback plan, environment and log review.

CHANGE REQUEST: IDEA -> DECISION -> AUDIT TRAIL

1 LOG > 2 ANALYZE > 3 APPROVE > 4 UPDATE > 5 VERIFY

Each request records: description; reason; affected deliverables; schedule / cost / security impact; approver; approval decision; baseline revision; test evidence.

WEEKLY RED / AMBER / GREEN REPORT

- **Schedule** — Due gates, sprint commitments, critical dependencies
- **Quality** — Pass/fail tests, known defects, user feedback
- **Risk & blockers** — New triggers, owner, mitigation, escalation
- **Budget / tools** — Model usage, deployment spend, resources

12 / LAUNCH & HANDOVER

Definition of done

The following acceptance contract is directly adapted from the source proposal's MVP criteria. Check each item only when demonstrated and backed by evidence.

- [ ] 01 Create a user project and complete at least four consecutive guided stages.
- [ ] 02 Enforce stage order and block implementation before required decisions.
- [ ] 03 Demonstrate at least three specialists using one shared project state.
- [ ] 04 Retrieve relevant technical knowledge from a vector-backed index.
- [ ] 05 Present two structured trade-off decisions with learner justification.
- [ ] 06 Generate at least two engineering artifacts from approved decisions.
- [ ] 07 Create/modify a code scaffold in isolation and execute a test/check.
- [ ] 08 Persist stage progress, decisions and artifacts across sessions.
- [ ] 09 Show completion and competency signals in the progress dashboard.
- [ ] 10 Deploy a working web app; hand over code, tests and clear README.

HANDOVER PACK

Project repo + commit history / source code + environment files / OpenAPI + diagrams / test reports + known limitations / deployment runbook / user guide / technical report + final presentation.

13 / TEMPLATES & PROVENANCE

Working controls

Use these registers from kickoff. Entries belong in the same project workspace as formal engineering decisions and artifacts.

RECORDS TO MAINTAIN

Decision log

ID | stage | alternatives | choice | justification | approver | timestamp

Risk / issue log

ID | description | impact | owner | response | target gate | status

Sprint board

Story | priority | owner | dependencies | definition of done | evidence

Artifact register

Type | project stage | version | generated_by | reviewer | approval

Change request

Request | impact | options | decision | approved_by | baseline version

BEFORE WORK BEGINS - UNBLOCK THESE DECISIONS

- Confirm supervisor, the four team members, kick-off date and final submission deadline.
- Approve the MVP agent set, model provider, hosting and monthly usage budget.
- Set rubric ownership, evaluation sample and numeric targets; confirm access to cloud / keys.
- Choose repository workflow, issue tracker and required review / merge policy.

SOURCE AND IMPLEMENTATION REFERENCES

Primary source: Model2Product AI - Graduation Project Proposal, August 2026, pp. 1-19.

FastAPI documentation
OWASP LLM Application Security
GitHub Actions